Lab 6 — Fit and Compress

Friday, December 4 (~20 min)

Last updated: October 29, 2026.

Friday, December 4 is **project launch** (capstone) with Lecture 40 (SVD) and Quiz 11. This lab is the same-day LS + truncated-SVD software warm-up — fine to meet SVD in lecture and practice it here. Quiz 11 is Week 13 eigenvalues — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Learning objectives

1. Solve a small least-squares / fit problem in software.
2. Run a truncated SVD on a small matrix and rebuild a rank-k approximation.
3. Compute residual size and storage count $k(m+n+1)$ for two values of k.

Math connections

Least squares (Lectures 38–39); SVD and the orthonormal / LS bridge (Lecture 40); warm-up for the image-compression capstone.

Key syntax (this lab)

Task	Octave	Python
Least squares	`A\b` (overdetermined)	`np.linalg.lstsq(A,b,rcond=None)`
Full SVD	`[U,S,V]=svd(A)`	`U,s,Vt=np.linalg.svd(A,full_matrices=False)`
Rank-k rebuild	`U(:,1:k)*S(1:k,1:k)*V(:,1:k)'`	`U[:,:k] @ np.diag(s[:k]) @ Vt[:k,:]`
Frobenius norm	`norm(A-Ak,'fro')`	`np.linalg.norm(A-Ak,'fro')`

Ops strip

Fit a line $y \approx c_1 + c_2 t$ to points $(t, y) = (0, 1)$, $(1, 2)$, $(2, 2)$.

C6.0. Build $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $b = (1, 2, 2)$. Solve for $\hat{c}$ with least squares (`A\b` / `lstsq`). Print $\hat{c}$ and the residual $\|A\hat{c} - b\|$.

Expected: $\hat{c} = (7/6, 1/2)$ (about $(1.167, 0.5)$); residual about 0.408.

C6.0b (bridge). Compute the SVD of this same A. Spot-check that the first two left singular vectors (columns of U) look orthonormal: products ≈ 0 off-diagonal, norms ≈ 1 (Lecture 14 vocabulary; Lecture 40 bridge). No need to rebuild $\hat{c}$ by hand from Σ — just notice SVD is the orthonormal toolkit behind a sturdy LS solve.

Expected: $u_i^{\mathsf{T}} u_j \approx 0$ $(i \neq j)$, $\|u_i\| \approx 1$.

Application

Work with the 4×4 matrix

$$M = \begin{bmatrix} 4 & 2 & 2 & 0 \\ 2 & 4 & 0 & 2 \\ 2 & 0 & 4 & 2 \\ 0 & 2 & 2 & 4 \end{bmatrix}.$$

C6.1. Compute the SVD of M. List the singular values (largest to smallest).

Expected: singular values $8, 4, 4, 0$ (or equivalent ordering with a zero).

C6.2. Build the rank-$k = 1$ approximation M_1 and the rank-$k = 2$ approximation M_2. Report $\|M - M_k\|_F$ for each k.

Expected: residual drops when k goes from 1 to 2 (exact numbers depend on rounding; $k = 2$ should be clearly smaller).

C6.3. Storage: for an $m \times n$ matrix, truncated SVD stores about $k(m + n + 1)$ numbers. For this 4×4 matrix, compute $k(m + n + 1)$ for $k = 1$ and $k = 2$. For a 100×100 grayscale image, which k values stay under half of $100 \cdot 100 = 10000$ entries?

Expected: here $k(4 + 4 + 1) = 9k$, so 9 and 18. For 100×100, $k(201) < 5000 \quad k \leq 24$.

One-pager

Preview the Lab 6 ticket; write on the hard copy in class.